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# Evaluating voting systems for Dutch national elections using machine learning

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# 1 Abstract

This thesis investigates the use of a Long Short-Term Memory neural network trained on Dutch election results, polling data, and voter survey data from the Dutch Parliamentary Election Studies (1989–2012) to predict party vote shares across four elections and analyse the effects of four alternative voting systems. The study compares the predictive performance and robustness of different dataset combinations using validation loss, seat error and robustness testing methods including noise injection, subsampling and feature ablation.

Results show that polling data contributes most strongly to predictive accuracy, while voter survey data primarily introduces noise, although income retains measurable predictive value. These predictions were subsequently applied to Condorcet, Borda count, ranked voting and Instant runoff voting (IRV). Across voting systems, the Condorcet method consistently selected VVD as winner, while Borda count and ranked voting favoured D66 or CDA, suggesting that centrist parties benefit disproportionately under preferential systems. Robustness analyses further showed that Condorcet was the most stable system under perturbation and bootstrap testing, whereas IRV and ranked systems were more sensitive to changes in voter preferences.

These findings demonstrate that voting system design meaningfully affects electoral outcomes and that machine learning can serve as a viable tool for evaluating democratic mechanisms.

# Contents

# 2 Introduction

'Voting can be regarded as a method of arriving at social choices derived from the preferences of individuals'
Kenneth Arrow [1]

## 2.1 Voting systems

Voting is what modern democracies are based on, but what is the best method? Fundamentally voting is the creation of a group decision based on the opinions of individuals within that group. There exists a wide variety of voting methods used around the world. The most used voting system globally, which is also used in the Netherlands, is plurality voting. In this system a voter can vote for a single candidate, the candidate with the most votes wins [2]. An example of a plurality election would be as follows.

| 50 | 40 | 60 |
|---|---|---|
| A | B | C |
| B | A | B |
| C | C | A |

In the columns the ranked preferences are shown, with on top how many people voted this way. Only the first row of the preference profile is considered for the vote, and so C would win with 60 votes.

A different voting method is the Condorcet method. After votes are cast the candidates are compared in head-to-head elections as shown in the tables below. The candidate which wins all the one-on-one elections is declared the winner [2]. An example of a Condorcet election goes as follows.

| 50 | 40 | 60 |
|---|---|---|
| A | B | B |
| B | A | A |

(a) A vs B

| 50 | 40 | 60 |
|---|---|---|
| B | B | C |
| C | C | B |

(b) B vs C

| 50 | 40 | 60 |
|---|---|---|
| A | A | C |
| C | C | A |

(c) A vs C

B wins its head-to-head with A and C, thus being declared the Condorcet winner.

Another method is instant run-off voting, which is a version of ranked voting used in Ireland and Australia. Voters rank the candidates in their preferential order. The candidate with the least first-choice votes is eliminated. This process continues until there is one winner remaining [3]. An example of an Instant Runoff election would go as follows.

| 50 | 40 | 60 |
|---|---|---|
| A | B | C |
| B | A | B |
| C | C | A |

In the first round, B would be eliminated, leaving A and C with

| 50 | 40 | 60 |
|---|---|---|
| A | A | C |
| C | C | A |

Here, A would be the winner with 90 votes.

Furthermore, there is score voting. It is similar to rank voting except for voters can assign a score to each candidate. If there are three candidates a voter would be able to give out a score of 3 points, 2 points and 1 point. The candidate with the highest score will be the winner [3]. Score voting is a form of Borda count voting, a system first devised in the year 1435 where every voter may give out scores starting with zero increasing until $n-1$ where $n$ is the amount of choices available. [4]. An example of a Borda Count election would go as follows.

| 50 | 40 | 60 | Score |
|---|---|---|---|
| A | B | C | 2 |
| B | A | B | 1 |
| C | C | A | 0 |

In this election, B would win with a total of 190 points as opposed to A with 140 points and C with 120 points.

In addition, there is the dictatorship in which, no matter what everyone else votes for, the dictator's vote wins. There are many more voting systems and different ways of implementing these voting systems.

Different modelling methods have been used to determine which voting method is the most fair or most acceptable to the largest group of voters. This led Kenneth Arrow to his impossibility theorem. In this he states that: 'If there are at least three alternatives among which members of society are free to order in any way, then every social welfare satisfying the conditions of transitivity, independence of irrelevant alternatives, and unanimity, must be a dictatorship' [1]. For example, in a situation where these conditions are met and a set of three voters need to choose between $A$, $B$ and $C$, the preferences may look like the following.

| Vote 1 | Vote 2 | Vote 3 |
|---|---|---|
| A | B | C |
| B | C | A |
| C | A | B |

If we look closely, the majority seem to prefer the choice $B$ over $C$ (voter 1 and 2), but the majority also prefers $C$ over $A$ (voter 2 and 3) and $A$ over $B$, (voter 1 and 3). This is exactly the paradox Arrow describes. Each voter has their own independent opinion, but not every voting system would give an equally satisfactory outcome for each voter. There is no clear winner using any mentioned voting method.

## 2.2 Voting in the Netherlands

This research will focus on Dutch national elections as there is publicly available data, like regularly taken polls and extensive voter surveys from the 1970's onwards. The national elections elect 150 parliamentarians by plurality voting. They are held every 4 years or earlier if the government falls and calls for new elections. Every vote cast is considered to be of one voting district which directly translates to the chosen candidate. There is no minimum percentage of votes that needs to be obtained. The number of votes needed to acquire a seat in parliament is determined by dividing the total people who voted by the 150 seats to be distributed. If, for example, a total of nine million valid ballots have been cast, the electoral quota will equal nine million divided by 150, i.e. 60,000 votes. This leads to a high number of parties in the Dutch parliament, as the bar to get in is low. In special cases, a parliamentarian might be elected by a preference vote. This entails that candidates who have received a total of 25% preference votes on top of the electoral quota are assured of a seat in any case. After this, the votes are distributed among the party's candidates in the order of their inclusion on the candidate list. Each party has a candidate list which includes all candidates per party in order of who will receive a seat once the threshold for another seat has been reached. The last remaining votes left over are divided among the parties by a set off complicated mathematical equations [5]. A full list of the party index used in this project can be found in table 20 in the appendix.

## 2.3 Machine learning

To analyse the data available on Dutch elections machine learning will be used in this project. Machine learning is increasingly used in all areas of research, including political and social sciences. Machine learning can be used to interpret big amounts exceeding human capacity. This can be applied to messages on social media, polling data and survey data [6][7][8][9]. Additionally, through its training mechanism it is suitable to make predictions based on these sets of data. This makes machine learning a good way of modelling the Dutch electorate for this research.

A type of machine learning model is the neural network (NN). Neural networks, sometimes also referred to as deep learning, are based on the connection between neurons in the human brain. By passing a set of known training data through the NN the model will learn to recognise patterns. This knowledge can then be used for new testing data [10].

An NN is made up of different layers, with each layer containing one or multiple neurons. Each neuron has a weight, this number transforms the input before it is fired to the next layer of neurons. Each neuron also has a bias which is added onto the input value. The NN is a web of neurons and by tweaking the weights and biases the desired outcome may be achieved. If the desired outcome is known this gets compared to the outcome made by the NN. The model will first choose the weights and biases randomly and then update them during the training to be closer to the expected or desired outcome by reasoning backwards [11].

## 2.4 Project aim

This research aims to answer the question; can conclusions be made about the application of different voting systems in the Netherlands based on machine learning research.

This project aims to broaden the research done on how machine learning can help us answer questions about the current voting system used in the Netherlands, how to improve on them and test their abilities. Models predicting elections have been made before but have not been used to attempt to analyse the many paradoxes present in political and voting theories. This will hopefully result in a method which can be used to get closer to answering the question of which voting system is the most representative, fair or even 'the best'.

# 3 Methodology

## 3.1 Materials

The electorate profile and their application to different voting systems will be done using machine learning. The specific type of neural network that will be used is a long short-term memory neural network (LSTM). LSTM is chosen specifically for its approach to time-series data. It is a recurrent network that uses both older (long-term) and more recent (short-term) data to make predictions about the future.

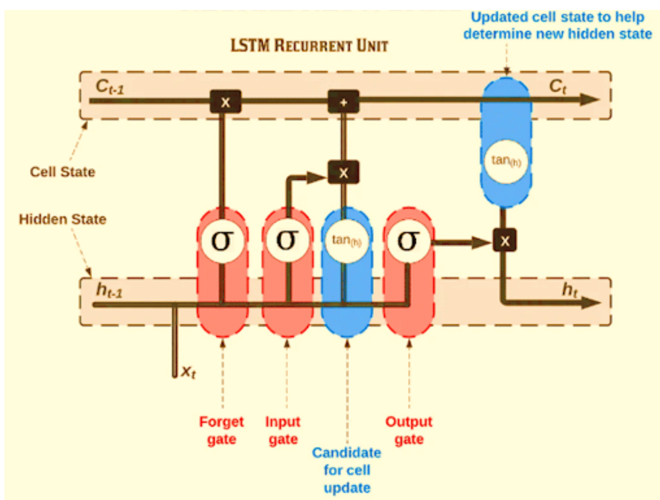


Figure 1: Overview of gates and states in an LSTM [12]

Figure 1 displays an overview of the different parts that make up a single LSTM cell. These cells are repeated for a certain sequence to train the prediction model.

The long-term memory, also called cell state, acts as the main information pathway through the network. Unlike a normal neural network layer, where information is repeatedly transformed by fixed weights, the cell state carries information forward across time steps with only limited modifications. This is how the LSTM can carry information early in a time-series forward through the training process, in contrast to regular recurrent normal networks where older information is more likely to vanish.

The prediction made by the LSTM is controlled by a set of gates, which in principle are a series of mathematical functions. The first gate is the forget gate, which decides how much of the previous cell state should be kept. The information is run through is sigmoid function. A result of zero means most information is removed and a result of one that most information is kept.

The second stage consists of the input gate and the candidate value. The input gate determines how much new information from the current input is added into the cell state, while the candidate value represents the new information that could potentially be added. Together, these components update the cell state by combining past information with relevant new information.

The updated cell state is then combined with the short-term memory, also called the hidden state. The hidden state represents the current output of the network at each time step and contains the information that is passed to the next stage and the output layer. Unlike the cell state, the hidden state is continuously modified by the network weights and reflects both the stored memory and the information currently considered important.

In the final stage, the output gate determines which parts of the updated cell state are put through as the hidden state and used for prediction. This allows the model to retain information internally without necessarily revealing it at every step. The process repeats over the full sequence of time-series data until a final prediction is reached [13][14][12].

For this project Python 3.12 was run on Pycharm. One of the most important libraries for deep learning is PyTorch, which transforms arrays into tensors. Tensors made in PyTorch are more suited to have mathematical operations performed on them and are optimised for deep learning purposes [15]. This will lead to faster runs needing less capacity. Additionally the Adman optimiser was used. This optimiser works on a gradient decent basis, which means it will try to find a minimum based on the gradients or slopes around that point. As the Adam optimiser is suitable for noisy and sparse data which fits with the used data sets [16]. The full code can be viewed and downloaded in GitHub [17].

## 3.2 Procedures

To fit the electorate data into this model survey questions will be turned into numerical values using one-hot-encoding. Vectors from different datasets will then be combined into an input matrix per election. The result will be given as a vector with the probability of voting for a certain party. This can then be transformed into an answer someone might give on a ballot in a specific voting system.

### 3.2.1 Data preparation

To create this electorate profile an array of data is used. This includes Dutch national election results, Dutch polling data by multiple polling institutions, data from the voter survey commissioned by the Dutch government after every election. Including not only voting data, but also census and economic data on the electorate will improve the completeness of the profile and enables the capturing of their influence on election outcomes [6].

After each election the Dutch government publishes the results on their website [18]. This data can be downloaded as a CSV file that includes the total votes per candidate and the total votes per party. Parties that changed their name or merged overtime were referred to by their most recent name. With this a party index was created that links the parties through the different datasets. To keep a concise overview of all of the parties they were encoded into a data frame [19]. From the original file the columns with the party names, the amount of seats and the amount of votes are kept. The percentage of votes and the percentage of seats are added in new columns. Everything was then combined into a matrix that can be put into the LSTM.

In the Netherlands there are an array of polling institutions. Polling data is collected and weighed by political scientist Tom Louwerse for the Peilingwijzer [20]. In the archive of the website an overview of all polling data since 1998 and a link to more detailed data on the Harvard Dataverse [21][22][23][24]. Tom Louwerse kindly shared the raw data belonging to the Peilingwijzer for this project which includes data from the following polling institutions: Ipsos I&O (since 2024), EenVandaag/Verian (since 2024), EenVandaag/GfK (2012-2019), I&O Research (2015-2024), Ipsos (earlier: Synovate/InterView) (1998-2024), Kantar (earlier: TNS Nipo/Kantar Public) (1998-2022), LISS Panel (2017), Peil.nl (2002-2019). To gather polling data from before 1998 data was scraped using ClaudeAI from allepeilingen.com as no raw data could be found [25]. This data is definitely less reliable, but was considered for use anyway as there were no other sources for this time period next to the election data.

To prepare the data percentages were all calculated on a 1-100 scale. Afterwards all data was sorted in chronological order and put into a matrix. For time periods where there were more than one source available, the preference was given to the polling data published by Tom Louwerse as the sources of his array of data are public.

Before and after each election the Dutch government sets out researchers to conduct a large voter survey. These surveys contain questions like, who did you vote for in the past, how satisfied are you with the current government and which other parties would you consider to vote for. This data gives an important inside into the thought process behind. Through nkodata.nl the datasets behind the surveys from 1989 until 2012 can be found [26][27][28][29][30][31][32]. Questions relevant for this study were picked on the basis that they say something about voting behaviour, thoughts on a political party or were census data. Additionally, only questions that appear in at least 3 surveys were considered. Each instance of the survey gave different variable names for the same questions. This was rectified for each question by hand. Questions were then split according to the answer model, binary, scaled or party. The parties were again encoded to ensure only numbers were input into the model. These values were then put into a matrix with their answer and timestamp for easy access and input.

### 3.2.2 The model

When running the code a choice can be made on which datasets can be used. Combinations can also be made, although the election results always need to be considered. The training/testing cut-off is made at 2012, thus 70 percent of the data is used for training the model and 30 percent for testing. The cut was made at a time stamp rather than randomly to make sure there was no temporal leakage. A time series was constructed for each political party and divided into fixed-length sequences of five consecutive observations. The model takes these sequences as input and attempts to predict the next value in the sequence. Each time step consists of a poll or election result concatenated with a vector of voter survey features representing voter attitudes at that point in time. The input dimensionality per time step is therefore $1 + k$, where $k$ is the number of retained voter features. For each sequence, five elements are stored and used throughout the training and evaluation process: the input sequence itself, the target value that the model must predict, the date associated with the target value, the seat count at the target date, and a boolean indicator specifying whether the target corresponds to an actual election result. The LSTM processes each sequence through a single hidden layer consisting of 64 hidden units. A fully connected linear layer then maps the final hidden state to a single scalar output representing the predicted vote share. The model was trained using the Adam optimiser with a learning rate of 0.005 and mean squared error (MSE) as the primary loss function. Training was performed for 250 epochs, which was found experimentally to provide the best balance between predictive performance and overfitting.

### 3.2.3 Application to voting systems

The predictions generated by the LSTM model are then combined with the most recent voter survey results to make ballots. The most recent voter survey that has published data is 2012. This means that all ballots are the same from 2012 onwards. The probability scores given by voters on which party they would vote for are used. These ballots are then used to determine the winners in different voting systems. The voting systems all have a function that can take ballot input and one that can take election results as input. The latter function is used to determine the 'actual' winner that is only based on the known election results and no other data. For the instant run-off voting some assumptions had to be made on implementation. Ties were broken by announcing the party with the most votes in the previous election as the winner and in the case of a true tie both parties were eliminated simultaneously. For the Condorcet method the assumption was made that in the case of a tie, neither vote was counted. For Borda count parties not present on a voter's ballot were not assigned any points.

## 3.3 Statistical analysis

### 3.3.1 Accuracy and loss

To describe the difference between the prediction make by a model and the real values accuracy and loss are the most common methods. The accuracy is simply the number of true predictions over the number of total predictions. The accuracy metric can describe how a model is performing at a specific moment.

The discrepancy between the predicted and actual value is called loss. The lower the loss value the less discrepancy [33]. There are two different types of loss at play here, the training loss and the validation loss. The training loss is solely based on the data used while training the model, while the validation loss looks at the loss of the predictions the model has made. The latter is most important to see how the model performs on future data while the training loss can indicate the efficiency of the training process. For each iteration of training or testing a loss can be determined. The development of loss over the epochs can be a valuable metric.

For the statistical analysis mean squared error (MSE), mean absolute error, the Huber loss function and the total seat discrepancy were used. The total seat discrepancy is the absolute sum of all positive and negative seat discrepancies per party.

MSE is the most commonly used method to compare predictions to actual values in cases of linear regression. The formula is as follows:

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2 \tag{1}$$

With the following variables:

- $n$ = total number of epochs
- $i$ = epoch
- $y$ = the actual value
- $\hat{f}(x_i)$ = the prediction given for the $i^{th}$ epoch

If the discrepancy between the actual and predicted value is small, the value for MSE will be as well [34]. When something is referred to as training loss or validation loss, unless specified otherwise, this is the MSE loss.

The mean absolute error (MAE) is also used in for linear regression models. More specifically, it often gets used over MSE if the model is resistant to a few outlying data points. The formula uses the same variables and is quite similar to that of MSE [33].

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{f}(x_i)| \tag{2}$$

The Huber loss function is a combination of the two functions above and specifically takes outliers into account.

$$\text{Huber loss} = \begin{cases} \frac{1}{2}(y - \hat{f}(x))^2, & \text{for}|y - \hat{f}(x)| \leq \varepsilon \\ \varepsilon\left(|y - \hat{f}(x)| - \frac{\varepsilon}{2}\right), & \text{otherwise} \end{cases} \tag{3}$$

In this research $\varepsilon = 1.0$, which is the threshold above which a data point is considered an outlier. The Huber loss is robust to outliers since the corresponding prediction errors $|y - \hat{f}(x)|$ are not squared. Outliers have a smaller effect on the average Huber loss over the entire dataset compared to MSE and MAE. Yet again, the smallest the value of the Huber loss the small the discrepancies between the predicted and actual value.

### 3.3.2 Robustness

Another important quality to test for this project is robustness. Robustness encompasses the stability of the model's results when inputs are altered. To test this, different methods were used.

Ablation in context of this project entails the leaving out of certain inputs. When testing the robustness of the LSTM this is only applicable if survey data is used. The different features within the survey data are taken out one by one to see which has the highest impact on the overall result. When applying the ablation method to the voting systems this was done by taking out one of the major parties. The five biggest parties over the timespan of this research are GL/PvdA, CDA, VVD, D66 and PVV.

Another robustness test is the adding of noise. Noise is any data that is not directly important to your model. The more noise is in the input the harder it becomes for the model to discern what is important and what is not. This can lead to worse performance [33]. Therefore it is a good metric to check the robustness of your model. When noise is added on purpose and this does not hugely impact the model, it can be considered robust. When the predictions do change drastically this means the model is margin dependent.

The size of the sampling data is also important to a model's robustness. When a model's performance stays stable when the amount of data put in drops the model is robust. The larger the sample size of a dataset the easier it is for the model to train to a high prediction accuracy [33].

For the application of voting systems specifically more specific testing was done. The input of ballots calls for different testing methods. Firstly an agreement check was done. A comparison was made between all voting systems and their predicted winners. The more the voting systems are in agreement with each other the more robust the model is.

To evaluate the robustness of the voting system outcomes under uncertain voter preferences, a voter perturbation procedure was applied to the generated ballot data. Perturbation analysis is commonly used in statistical modelling to examine the sensitivity of results to small variations in the underlying data, allowing the stability of a model or decision process to be assessed under noisy conditions [35]. In this case, perturbations were introduced by randomly flipping the top two preference scores of a subset of voters during each trial. This procedure simulates respondents who may have been uncertain between two closely preferred political parties when completing the survey. In case the system is robust, the flipping of the top two preferences should not lead to a change of winner.

Another commonly used statistical analysis is the bootstrap method. In this method the variability, and thus robustness, of a prediction can be determined. This is done through repeated resampling of a dataset, where a part of the current data gets replaced by other random data in the set. Through this, the goal is to approximate a result that plausibly could have come from the same dataset [34]. In this study, each bootstrap trial generated a new ballot dataset by randomly sampling voters with replacement from the original set of ballots while maintaining the original sample size. As a result, some voters may appear multiple times in a resampled dataset, while others may be deleted entirely.

Lastly, outcome differences were tested between different subgroups. Using the voter survey data subgroups were divided by idealogical position, education level, income and age. Each group was then divided into a low, medium and high group based on the answers given in the survey. The ballots of these subgroups were then run through the voting systems separately to determine their specific winner.

# 4 Results

## 4.1 LSTM

To test the impact of the different systems of the data set. testing was done on the LSTM that was trained on just the election results, the election result and the polling data, the election results and the voter survey and for all the data sets. This was done for 250 epochs, which was found through testing to be the optimal amount for usage of the model, and for 500 epochs to be able to see more results. Aditionally, the LSTM was tested for robustness through sub sampling, adding noise and abolition of features for runs that included survey data. Only the features which had a high impact on the model when removed are shown below.

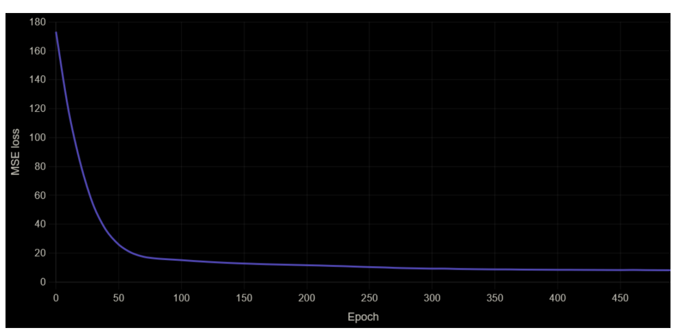


Figure 2: Mean squared error loss per epoch on election data to test the optimal amount of epochs

### 4.1.1 Election data

| | 250 epochs | | | | 500 epochs | | | |
|---|---|---|---|---|---|---|---|---|
| Date | MAE | MSE | Huber | Seat error | MAE | MSE | Huber | Seat error |
| 2012-09-12 | 4.200 | 69.048 | 3.784 | 38 | 6.221 | 62.569 | 5.804 | 56 |
| 2017-03-15 | 6.238 | 72.013 | 5.808 | 53 | 6.572 | 100.996 | 6.201 | 59 |
| 2021-03-17 | 7.692 | 100.553 | 7.259 | 68 | 7.930 | 141.534 | 7.499 | 72 |
| 2023-11-22 | 9.030 | 123.866 | 8.536 | 79 | 9.030 | 110.814 | 8.530 | 79 |

Table 2: Comparison of model performance for 250 and 500 training epochs

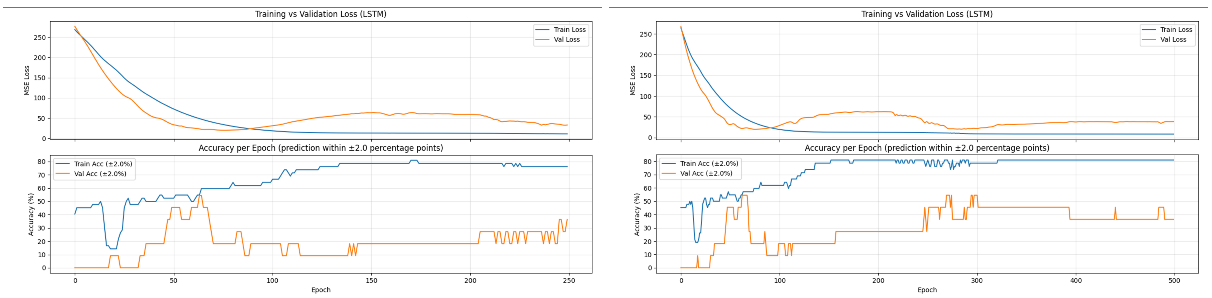


Figure 3: Loss and accuracy for 250 and 500 epochs

| **Subsampling Robustness** | | | |
|---|---|---|---|
| Fraction of validation set | Samples | Mean loss | Std. deviation |
| 50% | 10 | 58.2249 | 3.8859 |
| 70% | 10 | 58.1140 | 4.1051 |
| 90% | 10 | 57.8346 | 5.3629 |
| Baseline validation loss | 56.8447 | | |

| **Noise Robustness** | | |
|---|---|---|
| Noise level | Validation loss | Δ loss |
| 0.01 | 56.8762 | +0.0315 |
| 0.05 | 56.8486 | +0.0040 |
| 0.10 | 56.8863 | +0.0416 |
| Baseline validation loss | 56.8447 | |

Table 3: Robustness analysis of model

### 4.1.2 Election and polling data

| | 250 epochs | | | | 500 epochs | | | |
|---|---|---|---|---|---|---|---|---|
| Date | MAE | MSE | Huber | Seat error | MAE | MSE | Huber | Seat error |
| 2012-09-12 | 1.406 | 4.425 | 1.097 | 24 | 1.195 | 3.387 | 0.898 | 20 |
| 2017-03-15 | 1.286 | 3.272 | 0.895 | 19 | 1.286 | 3.383 | 0.910 | 21 |
| 2021-03-17 | 2.064 | 6.570 | 1.622 | 29 | 2.065 | 6.584 | 1.630 | 29 |
| 2023-11-22 | 2.591 | 14.156 | 2.178 | 40 | 2.614 | 14.724 | 2.201 | 40 |

Table 4: Comparison of model performance for 250 and 500 training epochs

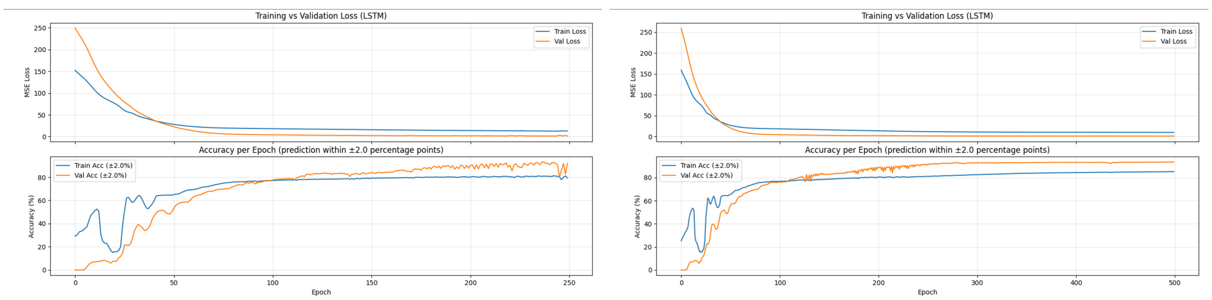


Figure 4: Loss and accuracy for 250 and 500 epochs

| **Subsampling Robustness** | | | |
|---|---|---|---|
| Fraction of validation set | Samples | Mean loss | Std. deviation |
| 50% | 1619 | 1.1287 | 0.0709 |
| 70% | 2267 | 1.1438 | 0.0415 |
| 90% | 2915 | 1.1178 | 0,0229 |
| Baseline validation loss | 1.1266 | | |

| **Noise Robustness** | | |
|---|---|---|
| Noise level | Validation loss | Δ loss |
| 0.01 | 1.1270 | +0.0003 |
| 0.05 | 1.1285 | +0.0018 |
| 0.10 | 1.1294 | +0.0028 |
| Baseline validation loss | 1.1266 | |

Table 5: Robustness analysis of model

### 4.1.3 Election and voter survey data

| | 250 epochs | | | | 500 epochs | | | |
|---|---|---|---|---|---|---|---|---|
| Date | MAE | MSE | Huber | Seat error | MAE | MSE | Huber | Seat error |
| 2012-09-12 | 7.538 | 75.583 | 7.038 | 68 | 6.270 | 65.872 | 5.770 | 58 |
| 2017-03-15 | 6.671 | 95.525 | 6.180 | 61 | 5.966 | 63.769 | 5.466 | 55 |
| 2021-03-17 | 8.189 | 125.109 | 7.689 | 72 | 7.539 | 88.323 | 7.039 | 66 |
| 2023-11-22 | 9.030 | 118.554 | 8.530 | 79 | 9.030 | 109.705 | 8.530 | 79 |

Table 6: Comparison of model performance for 250 and 500 training epochs

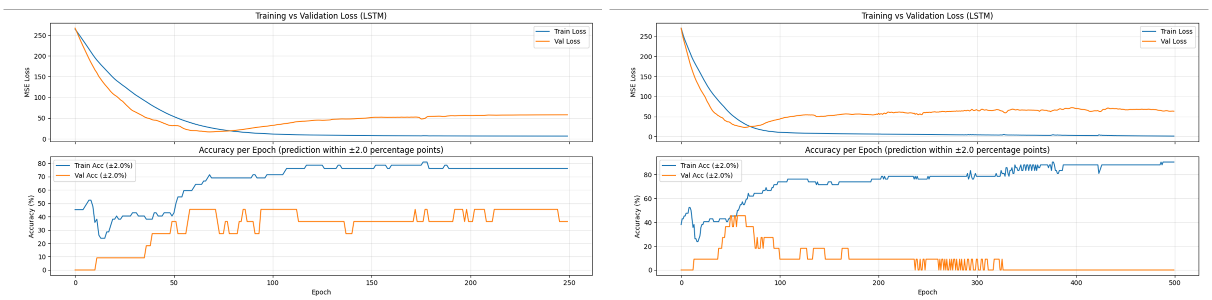


Figure 5: Loss and accuracy for 250 and 500 epochs

| Subsampling Robustness | | | |
|---|---|---|---|
| Fraction of validation set | Samples | Mean loss | Std. deviation |
| 50% | 10 | 53.7442 | 5.9663 |
| 70% | 10 | 53.7470 | 5.9092 |
| 90% | 10 | 51.5913 | 7.9661 |
| Baseline validation loss | 53.0448 | | |

| Noise Robustness | | |
|---|---|---|
| Noise level | Validation loss | Δ loss |
| 0.01 | 52.9878 | -0.0570 |
| 0.05 | 52.8639 | -0.1809 |
| 0.10 | 53.3718 | +0.03270 |
| Baseline validation loss | 53.0448 | |

| Feature Ablation | | |
|---|---|---|
| Feature name | Validation loss | Δ loss |
| poll_value | 200.0584 | +147.0137 |
| income | 56.7411 | +3.6963 |
| efficacy_external | 54.7082 | +1.6634 |
| left_right | 54.2509 | +1.2062 |
| satisfaction_gov | 54.1711 | +1.1263 |
| Baseline validation loss | 53.0448 | |

Table 7: Robustness analysis of model

#### 4.1.4 All data

| | 250 epochs | | | | 500 epochs | | | |
|---|---|---|---|---|---|---|---|---|
| Date | MAE | MSE | Huber | Seat error | MAE | MSE | Huber | Seat error |
| 2012-09-12 | 1.947 | 5.187 | 1.515 | 34 | 1.297 | 2.665 | 0.888 | 22 |
| 2017-03-15 | 2.056 | 6.910 | 1.597 | 31 | 1.938 | 5.991 | 1.492 | 29 |
| 2021-03-17 | 3.243 | 17.930 | 2.789 | 47 | 3.025 | 16.466 | 2.605 | 45 |
| 2023-11-22 | 3.392 | 22.420 | 2.925 | 50 | 3.432 | 25.106 | 2.964 | 52 |

Table 8: Comparison of model performance for 250 and 500 training epochs

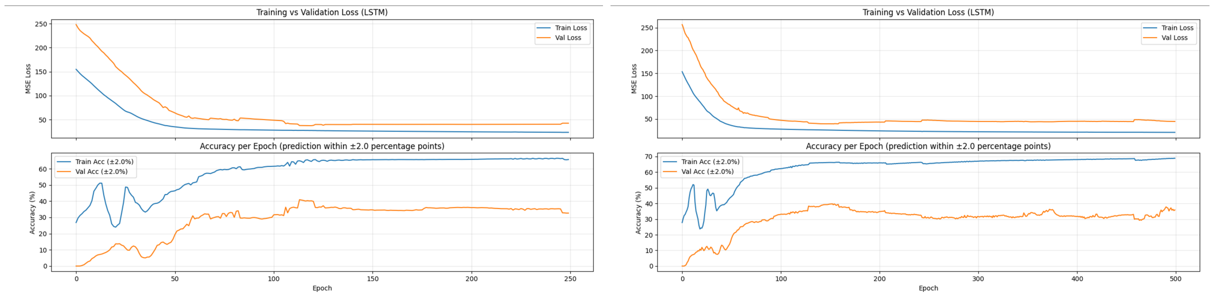


Figure 6: Loss and accuracy for 250 and 500 epochs

| **Subsampling Robustness** | | | |
|---|---|---|---|
| Fraction of validation set | Samples | Mean loss | Std. deviation |
| 50% | 1619 | 43.0280 | 1.1176 |
| 70% | 2267 | 43.0038 | 0.6642 |
| 90% | 2915 | 42.8859 | 0.4319 |
| Baseline validation loss | 42.8653 | | |

| **Noise Robustness** | | |
|---|---|---|
| Noise level | Validation loss | Δ loss |
| 0.01 | 42.8646 | -0.0007 |
| 0.05 | 42.8601 | -0.0053 |
| 0.10 | 42.8694 | +0.0041 |
| Baseline validation loss | 42.8653 | |

| **Feature Ablation** | | |
|---|---|---|
| Feature name | Validation loss | Δ loss |
| poll_value | 105.2268 | +62.3615 |
| income | 43.8968 | +1.0315 |
| Baseline validation loss | 53.0448 | |

Table 9: Robustness analysis of model

## 4.2 Application to voting systems

The voter survey data only extends to 2012. This means that the ballots constructed to represent the votes in different voting systems are the same each year. This leads to the predictions are the same for all combinations of datasets, which is why no breakdown per dataset is shown.

### 4.2.1 Predictions

| | Condorcet | | Borda | | Ranked | | IRV | |
|---|---|---|---|---|---|---|---|---|
| Date | Pred. | Actual | Pred. | Actual | Pred. | Actual | Pred. | Actual |
| 2012-09-12 | VVD | GL/PvdA | D66 | GL/PvdA | D66 | GL/PvdA | CDA | GL/PvdA |
| 2017-03-15 | VVD | VVD | CDA | VVD | CDA | VVD | CDA | VVD |
| 2021-03-17 | VVD | VVD | CDA | VVD | CDA | VVD | CDA | VVD |
| 2023-11-22 | VVD | PVV | D66 | PVV | CDA | PVV | CDA | PVV |

Table 10: Predicted and actual winners based on election data

### 4.2.2 Robustness

All tests were performed on the 2012 results.

| Voting system | Winner |
|---|---|
| Condorcet | VVD |
| Borda | D66 |
| Ranked voting | D66 |
| Instant Runoff (IRV) | CDA |
| 3 different winners detected | |

Table 11: Cross-system agreement of voting systems

| Voting system | Stability result |
|---|---|
| Condorcet | VVD (100%) |
| Borda | D66 (71%) |
| Ranked voting | D66 (69%) |
| Instant Runoff (IRV) | CDA (63%) |

Table 12: Ballot bootstrap robustness (100 trials)

| Voting system | Stability result |
|---|---|
| Condorcet | VVD (100%) |
| Borda | D66 (100%) |
| Ranked voting | D66 (100%) |
| Instant Runoff (IRV) | CDA (100%) |

Table 13: Voter perturbation robustness (100 trials)

| Removed party | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| GL/PvdA | Stable | Stable | Stable | Stable |
| VVD | Changed | Changed | Changed | Stable |
| PVV | Stable | Stable | Stable | Changed |
| CDA | Stable | Stable | Stable | Changed |
| D66 | Stable | Changed | Changed | Changed |

Table 14: Party ablation robustness

| Subgroup | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| Left | VVD | D66 | D66 | VVD |
| Centre | VVD | D66 | D66 | D66 |
| Right | VVD | D66 | D66 | CDA |

Table 15: Subgroup robustness by ideological position

| Subgroup | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| Low education | VVD | D66 | D66 | VVD |
| Medium education | VVD | D66 | D66 | VVD |
| High education | VVD | D66 | D66 | CDA |

Table 16: Subgroup robustness by education level

| Subgroup | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| Low income | VVD | D66 | D66 | CDA |
| Medium income | VVD | D66 | D66 | CDA |
| High income | VVD | CDA | CDA | CDA |

Table 17: Subgroup robustness by income

| Subgroup | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| Young | VVD | D66 | D66 | CDA |
| Middle-aged | VVD | D66 | D66 | CDA |
| Older | VVD | CDA | CDA | CDA |

Table 18: Subgroup robustness by age

| Subgroup | Condorcet | Borda | Ranked | IRV |
|---|---|---|---|---|
| Working class | VVD | D66 | D66 | CDA |
| Middle class | VVD | CDA | CDA | CDA |
| Upper class | VVD | D66 | CDA | CDA |

Table 19: Subgroup robustness by social class

# 5 Discussion

## 5.1 LSTM model

When looking at the differences in results on the different datasets a combination of election and polling data seems to be performing best. Datasets that do not include polling data perform with a way lower accuracy. This may be due to the polling data being the dataset with the most data points. Polling data has data points for every week, whereas the election data and voter survey only have data points each two to four years when elections happen. This also seems to reduce overfitting of the model as seen by the loss plots.

The figures of the accuracy of the runs not including any polling data (Figures 3 and 5) include bigger steps in accuracy that appear more square. This demonstrates the lack of data that can be taken in per epoch compared to the runs including the polling data. Because there is less data, the optimiser has to take bigger steps to try and find the optimal result.

The addition of voter survey data seems to only marginally improve the model's performance when comparing the results to only the election data. It decreases the loss slightly, but increases the seat error significantly. For the 2012 run the seat error based on only election data is 38 while this is 68 when voter survey data is added. Since this is the last set of survey data available chronologically, it is difficult to see the impact on the other testing years. However, it can be concluded that the voter survey answers mostly lead to extra noise and decrease the prediction's accuracy.

To test the impact of the different features within the voter survey data an ablation test was done for the datasets which included this data. It shows that the polling data has 60 to 100 times more impact than any of the single features within the survey data. This explains why the run with only election and polling data is more successful. It does not have the added noise of the survey data but does include the highly effective polling data. The feature that does seem to make the most impact on the validation loss is income. This endorses the idea presented earlier in this paper that census data will positively influence political prediction models [6].

## 5.2 Application to voting systems

The application of the LSTM to the different voting systems has as its biggest problem the lack of background data. The ballots, and thus the results, heavily rely on the probability scores from the voter survey. Since the voter survey of 2012 is the most recent instance of data that has been able to be found this means all the results presented as being from a certain date actually rely more on historical data than it seems at first glance.

Another important point to take into account is the presented 'actual' winners are only based on election results. In practice this means that for the ranking systems only the first choice was taken into account as other data is not available. This can be observed by the fact that the 'actual' values are the same across all voting systems. Even these 'actual' results give a skewed view of what the outcome might be when implemented in real life.

The Condorcet method seems to heavily favour the VVD. This may be the case because in during the tested time frame the VVD was always part of the coalition government and was the biggest party from 2010 until 2023. The VVD also worked together with a lot of different parties in these coalitions. Namely PVV, CDA, PvdA, CU and D66. Hypothetically, this enlarges the chance that someone who would vote for one of these parties also has a higher possibility of taking VVD into account in their rankings as their parties are not too far apart to be able to work together. This is reiterated by political scientist Tom van der Meer who argues that parties in the middle have become less distinguishable over the years [36].

Borda count and ranked voting determine either D66 or CDA to be the winner depending on the year. It makes sense that the results are alike as the voting methods are very similar. Only in 2023 do the results between the voting systems differ. This will be the effect of extra parties being added or older parties not existing any more creating a difference due to ranked voting giving one point even to the lowest rated party. In general it can be said that these two voting systems are more influenced by the probability scores as none of the winners are winners in the actual elections.

An earlier study showed that neural networks tend to the Borda count winner, even if trained otherwise [37][38]. This statement cannot be confirmed through this research. Firstly because the 'actual' Borda count winners are the same winners as in the other voting systems and therefore cannot be deemed representative. Additionally, there are only four total parties that emerge as an election winner. This pool is too small to conclude if this is because of the aforementioned effect or because these are historically the biggest parties in the Dutch system.

IRV always predicts CDA as the winner across all testing years. From this it can be concluded that the amount of parties with low to no votes do not impact IRV voting. This may be beneficial to the Dutch elections as a lot of small and new parties come up in every election [5].

## 5.3 Robustness testing

When comparing the robustness of the LSTM across the different combinations of datasets, it is noticeable that data sets that include polling data are more resistant to noise and subsampling.

Looking at the robustness tests done on the voting systems application, the choice was made to only run the tests on the 2012 results. This was done because this is the year on which the ballots are based and would therefore be the most representative result with the most up to date data.

Firstly considering cross-system agreement. Only Borda count and ranked voting agree on a winner. This is logical as the systems are closely related. The fact that the other voting systems show a different winner may not directly be because the testing is not robust enough, but simply because the margins in Dutch politics are relatively small.

The bootstrap robustness is very stable. The results show that for most voting systems aside from Condorcet there is a margin where the bootstrap does have some influence. As said before this may be due to the small margins and is unsurprising result.

As related to perturbation, it is surprising that this does not destabilise any of the results. This seems to imply that the first and second choices of most voters are relatively similar. This would mean that smaller rising parties will have a hard time getting through as a winner.

The ablation results indicate that the predicted election outcomes are most sensitive to the removal of VVD and D66 from the party set. In both cases, three out of four voting systems exhibit changes in their selected winners, suggesting that these two parties act as stabilising factors within the preference structure. In contrast, removing CDA has a more limited effect on Condorcet and Borda outcomes, which is notable since it appears as a winner more often than D66. This indicates that the preference for CDA is very dependent on which voting system is being used. This corresponds with the fact that CDA has a large voter potential but has not won a plurality election since 2006 [39].

Lastly, a couple of subgroups are used to test robustness. The differences between subgroups seem minimal, aside from the fact that CDA dominates in older and higher income populations. The Condorcet winner is still 100 percent stable. The fact mentioned earlier that income is the voter survey feature with the most impact can be seen in these results as well. CDA appears a lot when sorting people by income as a winner as seen in table 17 which coincides with the general results in table 10.

## 5.4 Research impact

This research contributes to the growing field of computational political forecasting by combining traditional election and polling data with voter survey data. While previous studies often focus either on forecasting election outcomes or on theoretical evaluations of voting systems separately, this thesis combines both approaches. Through this method, it is demonstrated that machine learning models can be used not only to estimate election outcomes, but also to explore how those outcomes may differ under alternative voting systems.

Additionally, this research also highlights the limitations of machine learning in the context of political sciences. The dependence on historical survey data and the lack of ranked ballot information lead to uncertain results when considering any other voting system than plurality. Nevertheless, this thesis provides a reproducible foundation for future research combining electoral forecasting, voter preference modelling and voting system analysis.

### 5.5 Further directions

There is an array of different directions future research could look into. Firstly, the incorporation of voter survey results can definitely be improved. In this research election and polling results were taken as the basic input and the voter survey only added later. This led to the fact that only the minimal amount of changes were made for the data to be incorporated. If the voter survey data was taken as the basis of which to make predictions, the model could give more insight into the effects different features have on the predictions.

Additionally, this model was made with predicting of seats and a winner in mind. The original plan was to make an electorate profile with broader attributes but this seemed beyond the scope of this project. It would of course still be interesting to explore a modelling of a Dutch electorate profile and the choices they might make in voting or other democratic processes.

Due to the abundance of polling data in comparison to other data this had a high influence on the results. With tweaking of weights and biases this could possibly be negated in the future.

Lastly, in the application of the model onto different voting systems only a single winner was presented. It would of course be a better comparison to the current plurality system if a seat division could be made. The extensive assumptions that would have to be made about the implementation of these voting systems makes this more social sciences research, which is beyond this project's goals.

## 6 Conclusion

This thesis explored the use of Long Short-Term Memory (LSTM) neural networks for predicting Dutch election outcomes and applying these predictions to alternative voting systems. Multiple combinations of election results, polling data and voter survey data were tested to determine which datasets contribute most effectively to accurate predictions and robustness.

The results show that polling data has the strongest positive effect on the predictive performance of the model. Combinations including polling data consistently achieved lower loss values, lower seat errors and smoother convergence during training. This is likely because polling data provides far more data points per time frame than election or voter survey data, allowing the model to better capture trends over time while reducing overfitting. In contrast, the addition of voter survey data only marginally improved prediction loss while often increasing seat error. This suggests that the survey data mainly introduces additional noise into the process. Nevertheless, the ablation tests demonstrate that certain demographic variables, especially income, still have measurable influence on prediction quality. This indicates that socioeconomic information may still be valuable when integrated more effectively into future models.

The application of the forecasting model to alternative voting systems shows that electoral outcomes can differ significantly depending on the voting method used. Across nearly all robustness tests, the

Condorcet method consistently selected VVD as the winner, while Borda count and ranked voting more frequently favoured D66 or CDA. This suggests that parties occupying central positions within the Dutch political landscape benefit from voting systems that reward broad acceptability rather than only plurality support. The implementation of these voting systems may therefore lead to a solidification of the centralists status quo.

At the same time, several limitations affect the interpretation of these findings. The ballots used for the voting system simulations heavily rely on the 2012 voter survey because no more recent survey data could be acquired. As a result, many of the presented outcomes are more dependent on historical preference structures than the election years themselves might suggest. Additionally, the "actual" election outcomes for the alternative voting systems are based only on plurality election results, since no real ranked ballots exist for comparison. This limits the extent to which conclusions can be drawn regarding how these systems would perform in practice.

However, the robustness tests provide important insight into the behaviour of the voting systems. Bootstrap resampling, perturbation analysis, subgroup testing and party ablation all show that some systems are more sensitive to small changes in voter preferences than others. Condorcet outcomes remained highly stable across nearly all tests, whereas IRV, Borda count and ranked voting showed greater instability under subsampling and ballot variation. This suggests that the Dutch political landscape contains relatively small margins between major parties, making some voting systems more vulnerable to slight shifts in voter rankings.

Future research could improve upon this work through the use of more extensive voter preference datasets, richer demographic modelling and simulations of proportional seat allocation under alternative voting systems.

# 7 Critical Reflection

During my thesis the overall theme of improvement was asking for help. While gathering data I was very reluctant to e-mail a professor from a different university who I knew would have access to the data I needed. This led to a delay in my work. I also had trouble asking for help when it came to making a realistic time planning. I would make them and then force myself to stick to them with no room for adaptation. This is something I am working on in therapy, but my lack of adaptation skills leads to a lot of delays. Another example of this is when I was gathering data by hand from a graph. I only decided that this was not a feasible option after spending two days entering 1400 data points into an excel sheet.

Furthermore a point of improvement is cancelling or postponing meetings. It is a reflex in cases of anxiety or the feeling I have not lived up to the goals I set for the week. Rather than discussing how I can improve on getting to my goals, or if they are even realistic, I prefer not asking for help and creating more time for myself. Ultimately leading to more delays cause I leave no room to adapt my planning.

Near the end of the project not asking for help was again a running theme. As the stress was getting to me more it was harder to keep up with basic human needs. This is something I have experienced before and know can be circumvented by asking for help. Due to shame, I am reluctant to ask for help.

Overall, what this thesis project has taught me is that I need more structure then I thought I did. Additionally, I should probably increase the frequency of my seeking (professional) help.

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# 9 Appendix

## 9.1 Party index

| Abbreviation / Name | Full Dutch name | English translation |
|---|---|---|
| 50PLUS | 50PLUS | 50PLUS |
| AOV | Algemeen Ouderen Verbond | General Elderly Alliance |
| AOV/Unie55+ | Algemeen Ouderen Verbond / Unie55+ | Elderly Alliance / Union 55+ |
| ARP | Anti-Revolutionaire Partij | Anti-Revolutionary Party |
| BBB | BoerBurgerBeweging | Farmer–Citizen Movement |
| BIJ1 | BIJ1 | As1 |
| BVNL | Belang van Nederland | Interest of the Netherlands |
| CD | Centrum Democraten | Centre Democrats |
| CDA | Christen Democratisch Appèl | Christian Democratic Appeal |
| CHU | Christelijk-Historische Unie | Christian Historical Union |
| CODE ORANJE | Code Oranje | Code Orange |
| CP | Centrumpartij | Centre Party |
| CP86 | Centrum Partij '86 | Centre Party '86 |
| CPN | Communistische Partij Nederland | Communist Party of the Netherlands |
| CU | ChristenUnie | Christian Union |
| D66 | Democraten 1966 | Democrats 1966 |
| DENK | DENK | THINK |
| DS70 | Democratisch Socialisten 1970 | Democratic Socialists 1970 |
| ETnNL | Eerlijk Transparant Nederland | Honest Transparent Netherlands |
| FvD | Forum voor Democratie | Forum for Democracy |
| GL/PvdA | GroenLinks / Partij van de Arbeid | GreenLeft / Labour Party |
| GPV | Gereformeerd Politiek Verbond | Reformed Political League |
| JA21 | JA21 | YES21 |
| JONG | Jong | Young |
| KVP | Katholieke Volkspartij | Catholic People's Party |
| LEF | LEF - Voor de Nieuwe Generatie | Courage - For the New Generation |
| LP | Libertarische Partij | Libertarian Party |
| LPF | Lijst Pim Fortuyn | Pim Fortuyn List |

| Abbreviation / Name | Full Dutch name | English translation |
|---|---|---|
| NIDA | NIDA | NIDA |
| NSC | Nieuw Sociaal Contract | New Social Contract |
| NVU | Nederlandse Volks-Unie | Dutch People's Union |
| PPR | Politieke Partij Radikalen | Political Party of Radicals |
| PSP | Pacifistisch Socialistische Partij | Pacifist Socialist Party |
| PVV | Partij voor de Vrijheid | Party for Freedom |
| PvdD | Partij voor de Dieren | Party for the Animals |
| PvdT | Partij van de Toekomst | Party of the Future |
| RPF | Reformatorische Politieke Federatie | Reformed Political Federation |
| SGP | Staatkundig Gereformeerde Partij | Reformed Political Party |
| SOPN | Soeverein Onafhankelijke Pioniers Nederland | Sovereign Independent Pioneers Netherlands |
| SP | Socialistische Partij | Socialist Party |
| Splinter | Splinter | Splinter |
| Volt | Volt Nederland | Volt Netherlands |
| VNL | VoorNederland | For the Netherlands |
| VVD | Volkspartij voor Vrijheid en Democratie | People's Party for Freedom and Democracy |

Table 20: Index of Dutch political parties used in the dataset.